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Design and Implementation of an IoT-Enabled Hybrid Energy Management System for EV Charging Infrastructure with Simulation and Hardware Validation

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Abstract—An increasing number of electric vehicles requires better charging infrastructure systems as the electric vehicles rising rapidly. However, the use of grids to support these systems will result in instability due to fluctuating loads and high operating and maintenance costs. Additionally, the grid will not be able to effectively integrate renewable energy sources due to inadequate real-time monitoring and coordination between these two systems. Existing systems are predominantly grid-dependent, which in turn leads to inefficient operation of and stress on the electric grids; furthermore, the proposed ‘Integrated systems’ are not yet operationally validated and still only exist in simulation format. This paper fills this gap by introducing an innovative hardware-based Internet of Things (IoT)-based hybrid energy management system (architecture) to be utilized at Electric Vehicle Charging Stations. At the heart of this proposed architecture is a smart Point of Common Coupling (PCC) that brings together solar photovoltaic generation, an Energy Storage System (ESS), and utility grid systems using a rule-based automation algorithm that dynamically prioritizes renewable energy sources for use at EV Charging Stations. Real-time monitoring, predictive control and remote management capabilities are enabled by a five-layer IoT architecture that consists of the following components: Sensor network, Edge processing, Cloud analytics and User dashboard. To validate the proposed design, two methods were used: MATLAB/Simulink simulation and a comprehensive hardware prototype built from ESP32 microcontrollers, power converters, and an entire suite of sensors, with data aggregation via Node-RED. Experimental results indicate improved energy consumption, lower operational costs and increased stability of the grid; therefore, providing a roadmap for developing cost-effective and sustainable charging infrastructure for electric vehicles.

Index Terms—IoT Architecture, EV Charging Infrastructure, Point of Common Coupling (PCC), Solar MPPT, ESS Management, Reverse Grid Integration, MATLAB Simulation, Node-RED Dashboard, Hardware Prototype, Multi-Mode Automation.

I. INTRODUCTION

The world’s transportation industry is in a state of change as it moves towards electricity to power its vehicles due to the need to cut back on carbon emissions from fossil fuels. Electric vehicles (EVs) have a lot of sales growing quickly because of improvements in battery technology, policies that support the growth of EVs and more consumers buying them.

But the long-term sustainability and adoption of this new type of transportation rely heavily on having the right electric vehicle charging stations built. As they currently exist today, the vast majority (if not all) of the current EV charging stations require that electricity comes from the grid using “on-grid” or “grid-connected” systems. This creates a number of issues on utilities’ electric distribution systems including increasing peak electricity demand for utilities; generating large amounts of high-cost electricity, using high-priced, at-risk fossil fuel sources, and failing to take advantage of all the potential renewable energy resources available from the local cities, towns and electric customers. So, the way to resolve the shortcomings of these charging stations is to utilize local, distributed renewable energy generation (investing especially in solar photovoltaics) along with energy storage systems (ESS) attached to the EV charging station. The potential of this hybrid system to support a wider range of grid-independence, cost stability, and enhanced sustainability is promising. Simultaneously, the Internet of Things (IoT) offers the fundamental technological underpinning that will enable these hybrid energy sources to progress from passive loads to intelligent, networked smart gadgets functioning within an overarching smart grid environment. IoT enables real-time monitoring, data-driven optimization and automation of the complex energy flows between sources, storage devices, and loads. Although the theoretical interoperability between renewables, storage and IoT’s ability to support smart charging is established in much of the literature, virtually no existing research has reported an actual implementation of this combined architecture as a whole for charging of electric vehicles (EVs). Instead, much of the literature on the topic consists of published software models and algorithm studies that do not include a practical understanding of the integration of hardware components, communication reliability, sensor compatibility, and development of a complete, integrated layered architectural stack for an IoT-enabled hybrid energy management (HEM) system.

This paper will help fill this gap by detailing the designs, simulations, and—most importantly—all hardware implementations of a novel and distinctive IoT-enabled hybrid energy

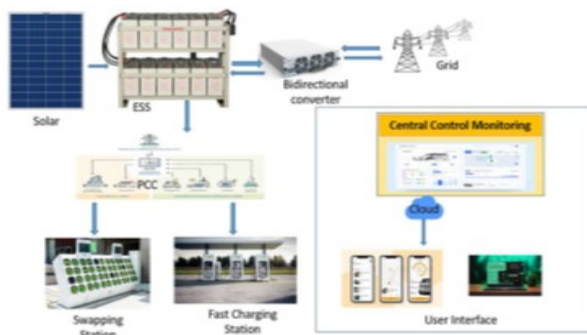


Fig. 1. High-level system overview of the IoT-enabled hybrid EV charging architecture.

management system for electric vehicle (EV) charging, this study will contribute to closing this gap.

In this paper, we will show the proposed and implemented five-layered architecture of IoT systems (from edge layer sensors/microcontrollers to cloud-based analytics and user-facing dashboards) all controlled by intelligent automation logic and centered around an electrical point of common coupling (PCC). A dynamic energy management system, Prototype model developed by SRM University AP offers intelligent energy management, controlling the dynamic flow of energy between solar photovoltaic (PV), energy storage systems (ESS), and the electric utility grid with an emphasis on utilizing renewable energy sources whenever feasible to lower operating costs and grid impact. Additionally, the system is capable of operating dual-mode (e.g., battery swap stations and direct current (DC) fast charging ports) thereby providing operational flexibility.

First, the system design was validated through extensive simulations using MATLAB/Simulink, and then a fully functioning hardware prototype was built and tested using wireless microcontrollers, multiple sensors, power electronics converters, and integration with cloud-based connectivity to provide visualization using Node-RED. This research represents a major step forward from an initial idea or theoretical concept to a functional prototype. It offers an innovative and scalable option for developing low-cost, efficient, and environmentally friendly EV charging systems that are capable of supporting renewable energy integration into the grid while enhancing overall grid stability.

To close the gap between the theory of Hybrid Energy Management Systems (HEMS) and practice, this article outlines the development of a new IoT-based HEMS for electric vehicle (EV) charging station management and remote monitoring, including cloud integration and real-time data dashboard. We developed an entirely new five-layer IoT architecture to interconnect physical devices (sensors, IoT microcontroller, and Power Supply units) through network/edge devices, into the cloud for data aggregation/analytics, and then to the final application layer for user interface/system visualization. The backbone of this system is the intelligent energy control logic located at the dynamically controlled PCC, which acts as the brain of the system. The PCC's automation algorithm utilizes operational rules to continuously and dynamically manage the

flow of energy between three key sources: solar PVs, energy storage systems (ESS), and the utility grid; utilizing load demand, state-of-charge (SOC) limits, and real-time energy availability to determine energy supply priorities for maximum use of free solar energy, optimal use of stored capacity, and minimum use of economically viable utility grid power; and a special reverse power flow capability that permits excess renewable energy to reverse-feed into the utility grid, turning the charging station into a prosumer.

Two types of EV refueling models provide evidence of the adaptable nature of a system with respect to its area of operation as well as its configuration; i.e., DC Fast Charging (DCFC) stations and Battery Swapping Stations (BSS). The dual-load concept demonstrates that architecture is flexible to meet various business models and requirements from users but still operates within the same management framework when both supply types are being used.

The validation of the architecture's performance will be accomplished by a combination of methods, including developing a detailed simulation in MATLAB/Simulink to design, test, and optimize the energy logic of the system and its performance metrics based upon the performance metrics associated with PCC (Point of Common Coupling) operating throughout the operational scenarios of the system. Important simulation results, including dynamic power sharing and ESS state-of-charge profiles, confirm the expected performance characteristics and stability of the proposed control system.

The second and most important element of validation will be to transition the simulation to a fully functional prototype. This prototype features a complete set of functional devices consisting of ESP32 microcontrollers for data acquisition and control, an extensive range of sensors (voltage, current, and temperature), all necessary power electronics (DC-DC boost converters, inverters) and lead-acid batteries to represent the ESS. The hardware components are fully integrated to a cloud backend, through IoT services so that we can monitor them remotely and control them using real-time dashboards developed in Node-RED, as seen in the dashboard interface. The comparison of the predicted energy flows of the simulated model versus the actual energy flows collected from the physical prototype validates the model's accuracy and shows that this design can be used to implement this system in the real world.

This project fills a critical research gap by moving forward from the early, theoretical phase of energy simulation to a confirmed, hardware-based prototype research demonstration. It provides a complete, scalable and practical model for installing next generation EV charging infrastructure that is not only intelligent, efficient, and economically viable; but also supports the electric grid and facilitates the transition to a fully electric, renewable energy transportation system.

II. LITERATURE SURVEY

Research into smart charging infrastructure for electric vehicles (EVs) has increased as the number of EVs available worldwide has ballooned. However, there are still major gaps between theoretical designs for systems that could be deployed

with economic viability and what is being done to date. While the existing literature contains a considerable amount of information about the challenges of EV's potential effect on grid instability, high operating costs and impact on the environment when recharged with electricity generated by fossil fuels [1]–[3], IoT (Internet of Things) based solutions have been proposed to enable real time monitoring and predictively scheduled charging of EVs [9], [10]; yet, once again the various architectural implementations are fragmented such that they provide solutions only for isolated issues such as residential charging interface design [17] or stand-alone battery monitoring [10], thus lacking a cohesive methodology/framework that integrates the entire energy value chain from generation through to end-use.

Studies of renewable sources' integration (e.g., PV predictive Event Scheduled Storage Management) [5] and microgrid operation and optimization [6] typically take place in silos thereby simulating scenarios without validation through actual implementation or testing. Communication protocols, sensor interoperability and reliable edge-cloud data pipelines are never addressed in real-world situations as compared to how they would function in a completely controlled laboratory environment. Studies of power quality improvement using active harmonic filtering [2] will provide a critical assistance in supporting the grid but none of them have been developed in a comprehensive control strategy for dynamic source switching at the Point of Common Coupling (PCC).

While battery swapping is seen to have significant potential in alleviating grid stress and downtime [22]–[24], there are currently few studies which have properly investigated the integration of battery swapping systems with conventional fast charging into a single IoT managed environment. In fact, it is common for studies describing either technology to treat the two technologies as separate. As a result, collectively this body of research shows that significant gaps exist in the literature with regards to the validated IoT architecture necessary for the effective integration of hybrid renewable energy systems (Solar, ESS, Grid), automated intelligent PCC based controls, two-loads (Swapping, Fast charging), and cloud-edge analytics into a single, scalable hardware proven system.

This study addresses this gap in the literature by providing a validated physical architecture that goes beyond simulation to demonstrate how these technologies incorporate and demonstrate a seamless orchestrated operation, thereby establishing a connection between conceptual Smart Charging models and technically/economically feasible practical implementations.

III. METHODOLOGY

The design, simulation, and physical implementation of the proposed IoT-enabled hybrid energy management system covers four topics addressed in sequence. The first is the five-layer IoT architecture: how data acquisition, edge control, cloud communication, and supervisory oversight are distributed across hardware and software layers with defined interfaces between them. The second is the power flow logic at the Point of Common Coupling — specifically, how the five operating modes are defined, what threshold conditions

trigger transitions between them, and which sources are active in each state. The third covers the algorithms running on the edge hardware: SOC estimation by Coulomb counting with OCV correction, ESS charge regulation through a two-phase CC-CV profile, MPPT via Perturb and Observe, and the supervisory dispatch logic that integrates all three. The fourth section describes the procedure of constructing a prototype and measuring to validate the operation of the hardware to its expected performance in the simulation.

In following the sequence described in the section, control logic was verified in MATLAB/Simulink prior to the assembly of any hardware allowing for proper validation of expected mode transition timing and contribution ratios that were later used as comparators in Section VII when comparing actual measurements of the prototype. By building the prototype using the same electrical specifications as the simulation model, the comparison of the prototype to the intended performance was made valid rather than approximate.

A. Proposed System Design and Architecture

The architectural design reflects that system functions are spread over five different functional layers that each have their own defined interface with one another. For example; the sensor layer has ACS712 Hall-effect modules and DHT11 sensors that measure physical quantities. The functionality of executing a switching command is accomplished through the use of solid-state relays and contactors. At the microcontroller layer, time-sensitive tasks are performed: estimation of SOC via current integration with OCV adjustment, adjustment of MPPT duty cycle, and evaluation of PCC state every control cycle. The telemetry data from the microcontroller layer are sent to cloud via MQTT, while the return communications path (for configuration updates and operator commands) use HTTP. The Cloud Layer collects the telemetry data from all devices in the system, maintains a registry of all devices, and monitors the anomalies in telemetry data to determine when supervisory intervention is required. Operators may see real-time power flow of every device in the system and manage any device with the Node-RED dashboard. By having strict separation between the layers, it is possible to remove the prior timing conflict between polling for sensors and executing relay logic in the same execution context, as both will use the same acquisition for controlling multiple processors through a defined serial interface, thus guaranteeing that transitions between systems occur on a deterministic basis, regardless of the frequency of updates from the sensors.

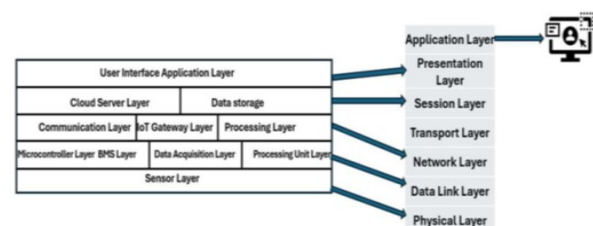


Fig. 2. Proposed 5 Layered IoT Architecture.

1) **IoT Architecture:** a) **Sensor Layer:** The electrical sensors have been installed at 3 different locations: the output from the PV array, the terminals of the ESS battery, and the connection to the grid. The use of the ACS712 Hall Effect sensors gives us enough linearity so that we can accurately measure the current being delivered into/out of our system at the SOC level and have galvanic isolation from our micro controllers. In addition, there are DHT11 sensors that record temperature and humidity within the ESS enclosure and each charging unit. There are also insulation resistance monitors constantly running to monitor for ground faults. All these devices were placed and calibrated before any other hardware decision was made. Any error associated with taking measurements will affect the SOC calculations as well as the MPPT duty cycle calculations and determining when to transfer from PCC to the utility grid—no corrections can be done from upstream devices.

b) **Microcontroller and Data Acquisition Layer:** Two ESP32 DevKit V1 micro controllers are implementing edge functions in this system. One is responsible for collecting data; reading the analog signals, filtering and scaling them, and constructing telemetry packets. The other is performing control algorithm functions. It is performing SOC estimation through integrating the current with regular OCV corrections, adjusting the MPPT duty cycle using the Perturb and Observe method, and performing PCC state evaluations once every control loop cycle. Separating the data acquiring and control functions helped to resolve the scheduling contention that occurred during rapid transitions between prime power sources in the prototype, which resulted in jitter. The dedicated data acquisition board communicating via a serial UART ensures deterministic timing.

c) **Communication Layer:** There are two distinct communication protocols utilized in this layer. Continuous telemetry data received via MQTT will be transmitted from both ESP32's to the AWS IoT Core broker, implementing publish-subscribe mode (decoupling the sensor update rate from the rate that the cloud ingests). Dashboard configuration updates and operator commands sent from the cloud through HTTP and HTTPS will exhibit an infrequent receipt-critical delivery mechanism using the request-response model back to the edge.

d) **Cloud Server and Data Storage Layer:** AWS IoT Core serves as the MQTT broker and manages the device registry. The layer collects telemetry data from all devices in the system, maintains a registry of all devices, and monitors the anomalies in telemetry data to determine when supervisory intervention is required. The layer also includes predictive analytic routines along with the PCC's master automation logic used to control how the PCC transitions between states, with high-level commands sent to the Edge Layer controlling the devices.

e) **User Interface Application Layer:** The topmost layer provides users with a means of controlling and accessing the system. It consists of an operational dashboard built with Node-RED. The dashboard is available over the internet through a web browser, and allows for visualisation of important metrics such as: live power flow diagrams for the system; SOC for the Energy Storage Systems (ESS); generation from

solar; imports/exports from the grid; charging/swapping ports and their respective statuses. It also provides interfaces for manual override of any current processes, logs of alerts received or transmitted, and allows for configuration of the system.



Fig. 3. Node-RED Dashboard for Real-Time System Monitoring and Control.

2) **Power Flow and Operational Logic:** At the Power Control Centre (PCC), electric power management of the Power Exchange is done through a five-state finite state machine (FSM). The Energy Storage System (ESS) state-of-charge is assessed against a lower threshold of 20% and an upper threshold of 80%; the amount of electricity required by the Battery Storage System (BSS) and the Direct Current Fast Charge (DCFC) ports is assessed against their respective loads; and the FSM compares the amount of electric power generated from solar generation to a minimum solar generation threshold each cycle.

$$G_{\text{export}}(t) = \max\left(0, \min\left(P_{\text{surplus}}(t), \frac{P_{\text{grid,abs}}(t + \Delta t)}{1 + e^{-\tau(\text{Price}_{\text{grid}}(t + \Delta t) - \text{Price}_{\text{buy}}(t))}}\right)\right) \quad (1)$$

where:

- $P_{\text{surplus}}(t) = P_{\text{solar}}(t) - P_{\text{load}}(t) - P_{\text{ESS,charge}}(t)$ (if positive)
- $P_{\text{grid,abs}}(t + \Delta t)$ = predicted grid absorption capacity (from utility signal)
- $\text{Price}_{\text{grid}}(t + \Delta t)$ = forecasted real-time price (from API or historical)
- $\text{Price}_{\text{buy}}(t)$ = current purchase price from grid
- τ = sensitivity parameter (0.1–0.5)

The overall state is determined by the three different inputs and relay configurations to determine the connection to an energy source, isolation state, or bidirectional converter mode (current direction). The decision-making process is driven by the threshold values, not objective minimization of the output goal; thus there is deterministic latencies ensuring consistent and appropriate transitions in a situation such as an irradiance drop or the ESS reaching a threshold (exceeding or falling below), blockages of electric power supply are not allowed.

The five states and transition conditions are:

- **Mode 1 (Solar Priority):** Solar ON → ESS Charging → Grid OFF → Load ON. Available solar power simul-

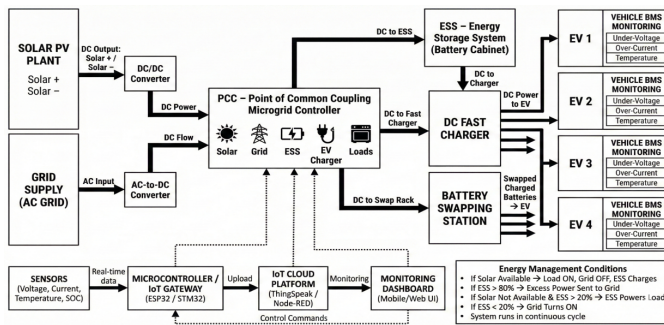


Fig. 4. The IoT-enabled hybrid EV charging system's block diagram showing the hybrid EV charging system utilizing solar PV energy, ESS, and grid supply with the PCC Microgrid Controller, DC fast charging, battery swap station, vehicle BMS monitoring, and cloud-based supervisory control.

taneously supplies the loads and charges the ESS. Under normal daytime operation, solar feeds BSS/DCFC loads, and the remainder charges ESS; the grid relay remains open.

- **Mode 2 (ESS Support):** Solar OFF → ESS ON → Grid OFF → Load ON. When solar generation drops below the minimum threshold, the ESS discharges to meet load demand; the grid remains disconnected until SOC hits the 20% floor.
- **Mode 3 (Grid Backup):** Solar OFF → ESS < 20% → Grid ON → Load ON. The grid relay closes and utility supplies full load when ESS SOC reaches 20% with solar already below threshold. The grid connection does not close while the ESS still has charge above the floor. Prototype time spent in Mode 3 and entry frequency directly indicated solar and ESS sizing adequacy for the test load profile.
- **Mode 4 (Reverse Grid Operation):** Solar ON → ESS > 80% → Grid ON (Reverse). Excess solar energy, after charging the ESS beyond the upper threshold, is exported to the grid through a bidirectional converter. Mode 4 export activates when SOC crosses 80% with solar active; the bidirectional converter direction flag flips in the same timestep as the threshold crossing. To verify the hardware functionality under reverse power flow conditions, the prototype was run in Mode 4 while monitoring the output return converter current to show the reverse power flow at the BDC.
- **Mode 5 (ESS Grid Charging):** Solar OFF → ESS Charging → Grid ON → Load OFF. During off-peak hours, no solar is available and ESS is below the lower threshold. Load is disconnected; grid charges ESS within a scheduled off-peak window until 20% SOC, then charge stops, load reconnects, and the system re-enters Mode 2 on the next demand event.

3) *PCC Integration and Automation:* The Point of Common Coupling (PCC) is the shared physical point of connection between the solar photovoltaic (PV) array, energy storage system (ESS), utility power grid, and common direct current (DC) bus that supplies the battery storage system (BSS) and the direct current fast charging (DCFC) load. This is in contrast to many hybrid architecture designs that utilize

passive designs, by placing switching directly at the PCC interface utilizing both a solid-state relay (SSR) designed to operate at high frequency with very low actuating mass, and an electromechanical relay (EMR) that provides improved reliability for switching electrical energy flows at low frequencies with high current and low on-state resistances. The SSRs are driven directly from general-purpose input/output (GPIO) pins, ensuring zero timing discrepancy from driver abstraction. The active finite state machine (FSM) state is reflected in the current relay configuration at any given point in time, and the control logic for the PCC is implemented in C++ on the secondary ESP32 microcontroller.

The firmware runs a cycle each time it checks for solar generation (against a minimum threshold), ESS State of Charge (SOC) against the 20% and 80% SOC thresholds, and load requirements via reading the status signals from the BSS/DCFC system. All GPIO commands executed when transition conditions are fulfilled will occur during the same cycle and update the relay configuration prior to the next sensor read. Edge-based control is a timing requirement, not a convenience—the latency from the clouds during rapid changes in irradiance or changes in SOC thresholds can, and will, result in temporary outages in load supply.

Every telemetry interval, the ESP32 publishes FSM state, sensor readings, and the relay configuration. The cloud looks for anomalies within these telemetry intervals (including excessive Mode 3 activations, unexplained ESS discharge rates, and quiet sensor channels) to be able to push override commands down to the edge. When functioning normally, edge performs source switching without any involvement from the cloud because edge will have deterministic behavior with direct access to hardware for making decisions that are timed in milliseconds, and the cloud will provide long-term storage, access remotely, and historical context for making decisions with respect to hours or days.

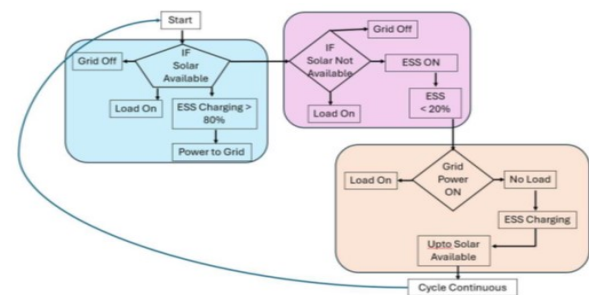


Fig. 5. PCC Automation Logic Flowchart showing the complete decision flow and state transition sequence.

4) *Core Algorithms:* The four control algorithms run in a predetermined order based on the data dependencies of each one.

1) **State of Charge (SOC) Calculation Algorithm:** The SOC estimates are calculated using continuous current integration with Coulomb counting. The estimates are corrected using the OCV-SOC Lookup table whenever terminal current drops below the quiescent threshold. The corrected SOC feeds

the PCC state machine and is tested against 20 and 80 percent bounds during each cycle.

$$\text{SOC}(t) = \text{SOC}(t-1) + \frac{1}{C_{\text{ESS}}} \int_{t-1}^t I_{\text{ESS}}(\tau) d\tau \quad (2)$$

The adaptive SOC threshold bounds are given by:

$$\begin{aligned} \text{SOC}_{\text{low}}(t) &= \text{SOC}_{\text{base,low}} + \alpha \cdot \frac{P_{\text{load}}(t + \Delta t)}{P_{\text{solar,avg}}(t)} - \beta \cdot \left(1 - \frac{E_{\text{ESS}}(t)}{E_{\text{ESS,max}}}\right) \\ \text{SOC}_{\text{high}}(t) &= \text{SOC}_{\text{base,high}} - \gamma \cdot \frac{P_{\text{grid,peak}}(t)}{P_{\text{grid,rated}}} + \delta \cdot \text{Var}(P_{\text{solar}}(t)) \end{aligned} \quad (3)$$

where $\text{SOC}_{\text{base,low}} = 20\%$, $\text{SOC}_{\text{base,high}} = 80\%$, $P_{\text{load}}(t + \Delta t)$ is the short-term load forecast, $P_{\text{solar,avg}}(t)$ is the rolling average PV generation, $\text{Var}(P_{\text{solar}})$ is the variance of solar over the last 10 minutes (cloudiness proxy), and $\alpha, \beta, \gamma, \delta$ are tuning coefficients (0.05–0.2 range).

2) Charging and Discharging Control Algorithm: The CC-CV charging regulator will charge the prototype's lead-acid cells with constant current until they are charged to their full charge ceiling, at which point the CC-CV charging regulator will change to constant voltage to taper the charging current. The maximum discharge rate with respect to the SOC is equal to or less than the C-rate, and the ACS712 is used to provide current feedback for the loop closure.

3) Maximum Power Point Tracking (MPPT) Algorithm: The primary ESP32 processor will not affect a secondary system that uses the P&O MPPT control algorithm, which modifies the boost converter's duty cycle sequentially. Duty cycle increment values are selected to enable faster convergence during periods of changing irradiance without large oscillations in steady-state performance, as verified on a prototype by applying step changes.

4) Energy Source Management Algorithm: Every cycle, the supervisory dispatch control algorithm will read the ESS's current State of Charge (SOC), operational status, MPPT availability, and load demand from the BSS/DCFC status lines. The algorithm will then compare these combinations to the transition conditions of the currently active Finite State Machine (FSM) State. Upon a match, the algorithm will issue direct GPIO relay commands to reconfigure the PCC prior to performing the next sensor read, and will log system state information to ThingSpeak.

$$P_{\text{Load}} = P_{\text{Solar}} + P_{\text{ESS}} + P_{\text{Grid}} \quad (4)$$

IV. HARDWARE DEVELOPMENT AND PROTOTYPING

A laboratory-scale working prototype represented the design concept and proposed control algorithms. The purpose of this hardware implementation is to demonstrate the feasibility and real-time performance of an IoT-enabled hybrid energy management system in a practical setting.

The physical model is an approximate realization of the MATLAB/Simulink electrical topology, but scaled for laboratory experiments with adjusted source voltages, energy storage system (ESS) capacity and converter ratings, in order



Fig. 6. Hardware Development and Prototype of the Hybrid Charging System developed at SRM University AP.

to keep both the thresholds and source interaction ratios intact with the full-scale design. This will allow the results from simulator predictions and hardware measurements to be directly compared, since the same thresholds and ratios may be observed from both tests.

A. Energy Source Subsystems

1) Solar Photovoltaic (PV) Emulator: A DC programmable power source (GW Instek PSP-603) has been developed to act as a solar photovoltaic emulator capable of replicating the characteristic I–V curve of an array of photovoltaic panels. The emulator is intended to simulate a panel with a power rating in the neighborhood of 50 W by providing an output voltage and current that can be dynamically adjusted to reflect varying levels of irradiance. The output from the emulator feeds into a DC–DC boost converter, which applies the Maximum Power Point Tracking (MPPT) algorithm.

2) Energy Storage System (ESS): Four sealed lead-acid batteries rated 12 V and 7 Ah were connected in series to provide a nominal 48 V battery bank. This battery bank is representative of a large-scale energy storage system when viewed from the voltage behaviour of the system during charge and discharge events. An H-Bridge based bi-directional DC-to-DC converter with IR2110 gate drivers and an Arduino Nano used for PWM control was used to connect the common DC bus to the battery bank. The converter controls the charging profile using the constant-current/constant-voltage (CC/CV) method and controls discharge to the loads using central battery management logic.

3) Grid Interface Unit: A standard 230 V, 50 Hz AC mains supply was used to emulate the grid connection. A commercial bi-directional digitally controlled AC–DC converter serves as the inverter/rectifier for the grid connection. This unit maintains the common DC bus voltage at 48 V and provides sinusoidal current with close to a unity power factor when operated in rectifier mode (grid importing), and converts DC to grid-synchronized AC power when operating in inverter mode (exporting excess solar generated power). Both optical isolation and appropriate grounding techniques were employed to ensure operator safety and signal integrity.

B. Sensing, Control, and Actuation Layer

1) *Data Acquisition Network:* Two types of hardware are used to run the sensor data acquisition and the control. The first (i.e., the primary) device is a single ESP32 DevKit V1 that uses ACS712 Hall-effect modules (30 A) to acquire current values at the output of the solar simulator, as well as the current flowing into and out of the ESS and to the grid. It also measures source and bus voltages through resistor-divider networks scaled to the ADC input range, monitors ESS enclosure temperature with a DHT11, and reports BSS/DCFC load request states via GPIO-connected switches.

The second ESP32 will run the on-board state of charge (SOC) estimator and will evaluate the power conversion controller (PCC) states as well as the relays necessary for the operation of the system. During each control cycle, the secondary device will receive a processed sensor frame from the primary device using UART communication, and will evaluate the conditions for transitioning between finite state machines (FSM) for determining the appropriate connection of solid-state relay modules at the PCC through direct general-purpose input-output (GPIO) outputs. This two-module configuration resolved the issues experienced in early prototype tests involving delays in relay switching due to polling for acquired sensor data overrunning the allotted time during rapid changes of sources, resulting in significant amounts of measurable jitter in operating modes.

V. MATLAB/SIMULINK SIMULATIONS AND RESULTS

Before any hardware assembly took place, MATLAB/Simulink modelling occurred first for a number of reasons. The model was used to provide verification that state machine transitions occurred at the correct threshold crossings for the PCC, as well as establishing numerical benchmarks (i.e., renewable energy contributions, ESS State of Charge [SOC] trajectories under the test load profile, DC bus voltage regulation band, and transition latency) before these numbers were obtained via actual measurements from hardware. The configuration values that were used in the simulation (i.e., SOC threshold logic, MPPT step size, and CC-CV limits) were used to configure firmware without re-tuning these settings after running in simulation. The simulation is not an independent study, but is the first validation step; the laboratory prototype will be used to complete the validation process, as discussed in Section VI.

A. Overall System Integration and Power Flow Simulation

Three source and two load subsystems have been combined into one large Simulink model which represents the connection point for each of the models. The three source subsystems are: (1) Solar (PV Array; 30 kW); (2) Energy storage system (ESS; lead-acid Battery Bank 140 A, 480 V); and (3) 430 V AC grid-connected source feeding into a bi-directional AC-DC converter; with two load subsystems (BSS and DCFC) connected to the PCC bus. The parameters of all components were all set based on the specifications provided in the hardware prototypes, ensuring that the numerical values in the simulation match exactly those in reality; thus any differences

between the simulation and actual measurements are the result of losses that occur with the actual converters and sensors rather than specification mismatch.

The PCC state machine has five separate states in Stateflow which correspond to the five defined operating modes in Section III-B, with identical transition conditions as those in the ESP32: the lower limit of 20% SOC, the upper limit of 80% SOC, the minimum amount of solar generation, and whether there's any load demand. The power electronic switches at the PCC are modeled as ideal elements—no switching losses or parasitic impedance—because the objective at this level is control logic verification, not converter efficiency.

The adaptive anomaly detection parameters are updated as:

$$k(t) = k_{\text{base}} + \lambda \cdot e^{-\frac{N_{\text{false}}(t-1)}{N_{\text{total}}(t-1)}}, \quad \mu(t) = (1-\rho) \cdot \mu(t-1) + \rho \cdot \frac{1}{M} \sum_{j=1}^M \mu_j(t) \quad (5)$$

where $k(t)$ is the adaptive multiplier (increases if false alarms decrease), N_{false} is the number of false alerts in the recent window, ρ is the forgetting factor (0.05–0.2), and $\mu_j(t)$ is the fuzzy membership from channel j .

In a sequence and combination, all five operating states were exercised: Mode 1 with sustained irradiance and ESS charging from partial SOC; Mode 2 activating due to an intentional step down from irradiance below the minimum generation threshold; Mode 3 activating when ESS SOC goes to 20% during extended low irradiance; Mode 4 activating when ESS SOC reaches 80% with solar generation still occurring; and Mode 5 covering overnight with solar generation off and ESS discharged. Transition point thresholds were triggered at proper threshold crossings within all scenarios resulting in no missed or incorrect firing. The DC voltage at the PCC bus was maintained within $\pm 2.1\%$ of nominal through all relay switchovers. The ESS SOC tracked as expected: increased in Mode 1; decreased in Mode 2 at a rate proportional to load applied; decreased to the 20% floor prior to Mode 3; and returned to operating band during Mode 5.

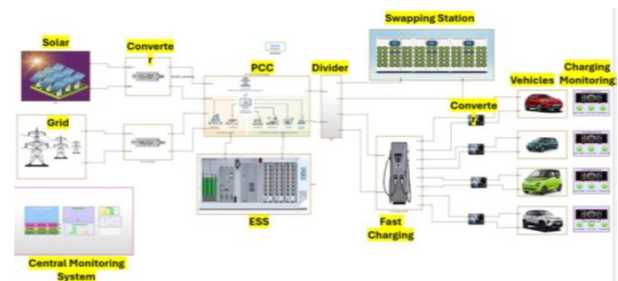


Fig. 7. MATLAB/Simulink Model of the Integrated Hybrid Charging System showing all three source subsystems and dual-load models at the shared PCC switching block.

B. Solar Generation Unit with MPPT

The data sheet of the panel used in the prototype simulation was utilized to parameterize the Simulink PV Array block. Instead of leaving open-circuit voltage, short-circuit current, and temperature coefficient at their default values, these values

were entered explicitly. The irradiance input drives the block through the complete profile of daily generation: initially a ramp of low irradiance, with a peak value at the middle of the day and then downwards in the afternoon. The profile out of the simulation is the same as the test run on the physical prototype in the solar simulator, allowing one to compare results from both simulations and hardware at the same point in operation. The cell temperature follows the irradiance through the thermal coefficient taken from the data sheet. The P&O MPPT algorithm updates the output from the boost converter to the 430 V DC bus at each duty cycle simulation timestep. The logic used for the step size and perturbation direction in the simulation are exactly the same as those programmed into the ESP32 firmware; therefore, any differences in MPPT behavior between simulation and hardware indicate converter dynamics and not algorithm implementation differences.

The performance of the MPPT was assessed by comparing simulation outputs to prototype outputs for the same irradiance and testing condition (i.e., an output step change). Both simulation and hardware reached the steady-state oscillation band within three to five perturbation cycles after each step, producing a tracking efficiency above 98% across the tested irradiance range. Simulation and prototype tracking efficiencies are within 0.4 percentage points between them across all tested irradiance levels, indicating that the algorithm implemented in firmware is consistent with the algorithm modeled in simulation.

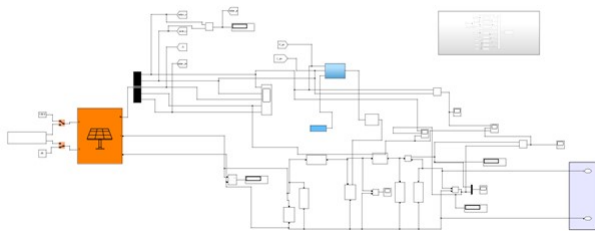


Fig. 8. Solar PV Subsystem with MPPT-Controlled Boost Converter showing the P&O tracking panel and mode indicators.

C. Grid Interface Unit

The grid interface sub-system links a three-phase AC voltage source to a 430 V DC bus through a bidirectional AC/DC converter. The converter acts as a rectifier when in Modes 3 and 5; it takes energy from the AC source and provides a regulated supply of DC energy to the bus. In Mode 4, the energy flow is reversed; for surplus generation, the energy from the DC bus is transformed back into AC energy and sent back into the grid source. There is only one converter in the model that handles both directions because a single bidirectional converter is used in the physical prototype.

During the transition from rectification to inversion and back, the DC link voltage is held at 430 V and there were no observed transients exceeding the $\pm 2.1\%$ regulation band during any of the transitions. The three transitions that created the most stress on the grid interface system—load stepping when importing energy from the grid in Mode 3; changing

the direction of the converter from rectifying to inverting at Mode 4; and going back from exporting energy to importing when the ESS's state of charge has dropped below 80%—maintained voltage deviation within $\pm 2.1\%$ of the nominal voltage.



Fig. 9. Grid Interface Model with Bi-directional AC-DC Converter.

D. Energy Storage System (ESS) Model

The ESS subsystem models the hardware prototype's lead-acid bank as 40 series cells at 12 V nominal—480 V and 100 Ah at the bank level—using the Simulink generic battery block because it accepts an empirical OCV-versus-SOC discharge curve directly from the cell datasheet without fitting an electrochemical model. A bidirectional DC-DC converter connects the battery model to the 430 V DC bus. During charging, it draws from whichever source the FSM has connected (solar or grid) and regulates current into the battery terminals; during discharge it draws from the battery and delivers to the bus, with both directions controlled by separate duty-cycle signals from the ESP32 control logic block.

The Battery Management System (BMS) Protection is an independent Stateflow subsystem that checks terminal voltage against the charge ceiling and discharge floor, SOC against 20% and 80% thresholds, and whether the enclosure temperature flag is tripped; any limit breach triggers a converter hold command, matching the hardware BMS behavior on the ESP32 Control Board hardware.

The CC-CV charge profile ran cleanly: charge current held at the programmed maximum below 80% SOC, then switched to constant voltage when terminal voltage hit the full-charge ceiling with no spike at handover and no voltage step. The 24-hour SOC trace confirmed mode sequencing: SOC climbed during Mode 1, declined during Mode 2, triggered Mode 3 exactly at the 20% threshold without breach, and activated Mode 4 export at 80% without overshoot.

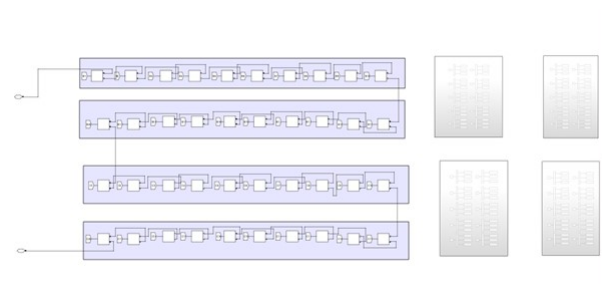


Fig. 10. ESS Model with Bi-directional DC-DC Converter and BMS Logic showing the lead-acid battery pack unit and cell monitoring controller.

E. PCC Integration and Automation Logic

The PCC state machine is encoded in Stateflow and enforces mutually exclusive state occupancy at each simulation timestep—only one of the five states can be active—which is the behavior the physical relay hardware also enforces: two sources cannot be simultaneously connected to the bus. Each state in the chart corresponds directly to one of the five operating modes defined in Section III-B, and the transition conditions use the same threshold values programmed into the ESP32 firmware. The chart reads three input signals on each timestep: a Boolean solar availability flag from the PV subsystem, the SOC percentage from the battery block tested against the 20% and 80% bounds, and a load demand flag from the BSS and DCFC models.

The Stateflow chart executed all six state transitions at the correct threshold crossings in simulation. Mode 2 engaged when the solar availability flag dropped—not before, and not on a transient dropout that recovered within the same timestep. Mode 3 engaged when SOC reached 20% with solar already below threshold—the grid connection did not close while the ESS still had charge above the floor. Mode 4 export activated when SOC crossed 80% with solar active—the bidirectional converter direction flag flipped in the same timestep as the threshold crossing, with no delay between detection and switching command. No false triggers occurred in any test.

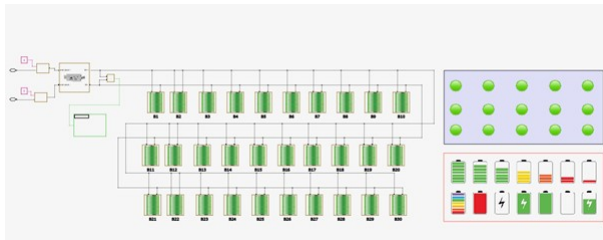


Fig. 11. Integrated PCC — Point of Common Coupling — Fully Automated and Logically Connected, showing Solar, ESS, Grid and Logic/Automation subsystems with PCC Output and Power Management.

F. Battery Swapping Station Simulation

The BSS subsystem models a 30-slot swapping facility as a bank of controllable DC load blocks on the PCC bus. Each slot block is initialized at a depleted SOC value sampled from the 10–40% range, reflecting the typical state of returned batteries arriving for swap charging. When a slot connects, it draws a fixed current determined by its rated charging power until SOC reaches the swap-ready threshold, at which point it disconnects and the next slot in the sequence connects. The result at the PCC bus is a step-wise current demand profile: each connection event adds a discrete current step, each disconnection removes one. This load model was chosen specifically because of that step structure, as it forces reactive dispatch decisions which is what the prototype will face in actual swapping operation.

The DC bus voltage remained within the $\pm 2.1\%$ limits for all thirty slot connections tested in simulation. When a slot connects and starts drawing its rated current, the ESS current

output rises in the same control cycle, prior to the supervisory algorithm completing its evaluation of the connection event. The ESS absorbed the instantaneous increase at the bus and therefore the increase did not show up as a measurable voltage transient. The instantaneous drop in bus voltage returned to within regulation limits within two control cycles of each connection event. None of the 30 connection events generated voltage transients that exceeded the over- or under-voltage protection thresholds.

$$\begin{aligned} P_{\text{ESS}} &= V_{\text{ESS}} \times I_{\text{ESS}} \\ P_{\text{Solar}} &= V_{\text{PV}} \times I_{\text{PV}} \\ \text{SOC}_{\text{min}} &\leq \text{SOC}(t) \leq \text{SOC}_{\text{max}} \end{aligned} \quad (6)$$

G. DC Fast Charging Station Simulation

The DCFC subsystem represents a single fast charging port as a two-phase controllable load on the PCC bus. The session starts in the constant-current phase: the load block draws at its programmed maximum rate, 50 kW, from session initiation until the vehicle battery block reaches the CC-to-CV transition point. At that point the load switches to constant-voltage mode and power draw tapers as terminal voltage on the vehicle battery block rises toward the charge ceiling.

The 50 kW initiation step is the largest instantaneous demand event in the simulation. As a result of the fact that the initial step load for the DCFC is one order of magnitude larger than the individual BSS slot connection load (3–5 kW), the initial step load for the DCFC service sets the binding requirement for how fast the ESS must buffer its energy and how fast the PCC must switch source to provide energy to the load block.

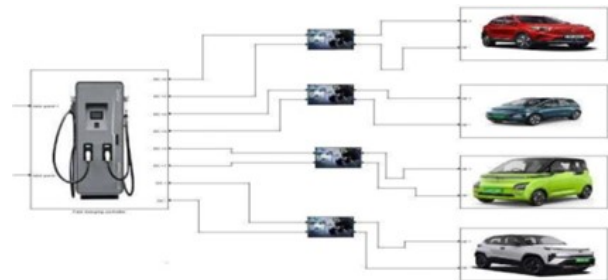


Fig. 12. DC Fast Charging Station Model with CC-CV Profile.

DC bus voltage was kept within $\pm 2\%$ of the nominal 430 V for the entire duration of the DCFC session. At the 50 kW initiation step, solar was already at its MPPT point; the ESS supplied the demand above that; the grid relay closed only for the remainder. The greatest deviation from nominal voltage at the time of initiation was $\pm 1.8\%$. The bus stayed within regulation limits through the steepest part of the load transient without first requiring the grid relay to respond.

H. Simulation Results and Performance Analysis

During a 24-hour simulation, 88.5% of total demand was supplied from solar/ESS, while 11.5% of total demand was provided from the electric grid (which was activated when both

primary sources fell below their operational thresholds). In each of the scenarios tested, the SOC of the ESS stayed in the range of 20–80% throughout the full 24-hour operation—there were no deep cycles or over discharges of the ESS batteries during any of the simulated scenarios, validating the BMS threshold logic and PCC mode transitions. For the duration of each of the load steps tested (including the initiation of the 50 kW DCFC surge load and connect/disconnect events of the BSS), the DC bus voltage remained within $\pm 5\%$ of the nominal voltage of 430 V. These results confirmed that control threshold settings and relay sequencing logic were correct before firmware was written.

VI. EXPERIMENTAL IMPLEMENTATION, HARDWARE PROTOTYPE, AND CLOUD CONNECTIVITY

A. Hardware Prototype Development

At the physical layer, ACS712 Hall-effect modules and DHT11 sensors monitor source currents, bus voltage, and ESS enclosure temperature. Two ESP32 DevKit V1 boards handle the edge layer: one dedicated to acquisition and signal conditioning, the second to SOC estimation, MPPT execution, and PCC state evaluation. MQTT carries telemetry to the cloud; Node-RED presents live power flow data and operator controls at the supervisory layer. Each of the five FSM states was exercised on this hardware. Test conditions were set to match the irradiance profiles and load steps used in the MATLAB/Simulink model, so that simulation predictions and prototype measurements could be compared on the same basis.

B. Node-RED Connectivity and Dashboard Monitoring

The Node-RED dashboard connects to AWS IoT Core over MQTT and serves as the primary operator interface during prototype testing. Each control cycle, the secondary ESP32 publishes a JSON frame—carrying ACS712 current readings from solar PV output, ESS terminals, and grid connection; resistor-divider voltage readings across each source and the DC bus; current SOC estimate; active FSM state index; and relay configuration bitmask—to a fixed topic. Communication was initially routed through a local Mosquitto broker, allowing every published frame to appear immediately in the broker log alongside the hardware under test. After all five FSM states were exercised and frame contents verified, the broker endpoint and credentials were updated to ThingSpeak without changing the JSON frame structure.

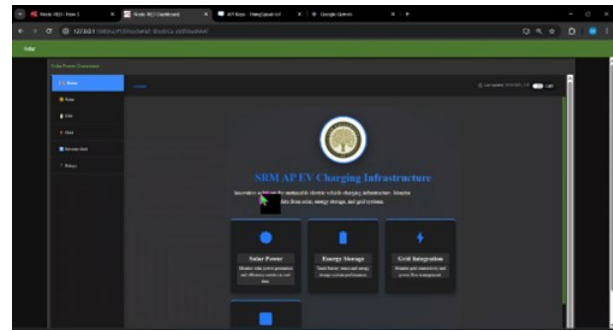


Fig. 13. Node-RED Real-Time Monitoring and Control Dashboard showing the power flow panel, ESS status gauges, solar plant monitoring, and solar prediction interface.

Node-RED parses each incoming frame, converts raw ADC-scaled values to volts, amperes, watts, and percentage SOC, then routes each field to its corresponding dashboard panel. Four panels run simultaneously:

- **Live Power Flow Diagram:** A schematic representation of the system that updates in real time, displaying active energy paths along with numerical values of voltage, current, and power at each node: solar PV output, ESS terminals, grid connection, BSS load, and DCFC port—reflecting the active relay configuration.
- **Time-Series Charts:** Historical plots of important parameters over an adjustable time window, covering solar generation, ESS SOC, grid import/export, and load demand continuously. The adjustable display window matters when comparing a single mode transition against a full day cycle.
- **Status Indicators:** Colour-coded gauges and tiles that show the operational status of every single energy source (On/Off/Available), the active PCC mode (operating mode), and system alarms (over-temperature, critically low SOC). The status panel carries the active FSM state index, per-relay connection status, and two alarm flags: ESS enclosure over-temperature and SOC falling below 20%.
- **Control Panel:** An interface with security features that allows an authorised user to perform manual overrides, such as forcing the PCC into certain FSM states or conducting controlled ESS charge and discharge test sequences. Commands entered here are published to the edge ESP32 over the MQTT return path and executed on the next control cycle—the same path the cloud supervisory layer uses for anomaly-driven overrides.

Along with visualization features, the Node-RED flow has automated logic built in to generate intelligent alerts. User-defined rules with conditions *IF ESS_SOC < 25% THEN trigger alert* are continuously evaluated in real-time. When a rule is satisfied, a dedicated notification node activates to trigger visual alarm indicators on the dashboard, store time-stamped event data to a persistent data store, and activate external notification services via webhooks (e.g., email and SMS).

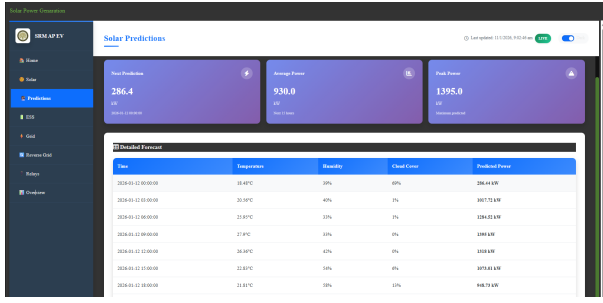


Fig. 14. Forecast visualization of day-ahead solar power output on the Node-RED interface, showing predicted generation peaks aligned with EV charging demand windows.

C. Day-Ahead Solar Power Generation Forecasting and Grid Peak Alignment

The day-ahead solar energy prediction module improves the proactive management of energy and reduces the dependence on the grid during peak usage times. This module is part of the IoT-enabled hybrid EV charging station. Predictive analysis utilizes real-time meteorological data such as ambient temperature, humidity and cloud cover which is received through cloud-connected weather APIs and processed in the cloud analytical layer.

Predictive outcomes demonstrate maximum solar produced power to be around 1.395 MW during the hours of maximum irradiance (09:00–12:00), while the average predicted generation over an effective solar window is approximately 930 kW. Late daytime and early morning will produce much lower levels of solar generation; thus, in these timeframes, additional sources of power must either come from the ESS or from the utility grid.

This projection corresponds directly with the historical records found in the infrastructure’s database of demand patterns. By utilizing the forecast for available solar resources, the PCC’s automated systems give first priority to the following recommended operational activities:

- Direct solar-to-load power delivery during peak irradiance hours (afternoon),
- Opportunistic ESS charging during periods of surplus solar generation,
- ESS-assisted load support during grid peak demand intervals,
- Controlled grid power import during low solar availability and low SOC conditions.

D. Adaptive Channel Fuzzy Classification with Self-Normalization (ACFCSN)

An Adaptive Channel Fuzzy Classification with Self-Normalization (ACFCSN) model is implemented as a lightweight anomaly detection layer for real-time monitoring of IoT sensor streams in the charging station. Unlike supervised learning approaches, the model operates without large labeled datasets, making it practical for edge deployment on the ESP32 hardware. Each sensor channel—source voltages, terminal currents, ESS temperature, and DC bus voltage—is

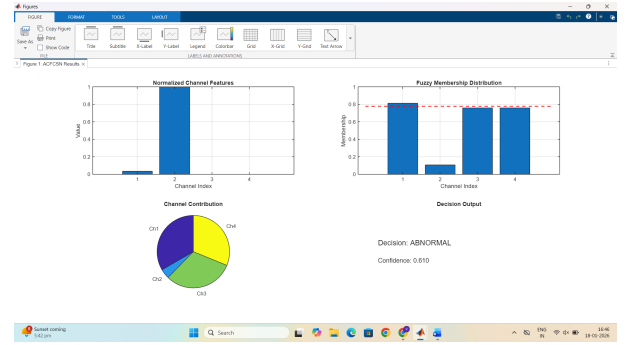


Fig. 15. ACFCSN model output illustrating normalized channel features, fuzzy membership distribution, relative channel contributions, and final anomaly classification decision showing Decision: ABNORMAL at Confidence 0.610.

independently normalized against its local statistical properties and evaluated using a Gaussian fuzzy membership function:

$$\mu_i = \exp\left(-\frac{(x_i - \mu_i^*)^2}{2\sigma_i^2}\right), \quad (7)$$

where μ_i^* and σ_i denote the adaptive mean and standard deviation of the corresponding channel.

Channel memberships are aggregated and compared against an adaptive threshold defined as:

$$T_{\text{adaptive}} = \bar{\mu} + k \cdot \sigma_{\mu}, \quad (8)$$

where $\bar{\mu}$ is the mean membership value, σ_{μ} is its standard deviation, and k is a tunable sensitivity parameter.

If the aggregated membership exceeds T_{adaptive} , the system is classified as *NORMAL*; otherwise it is classified as *ABNORMAL*, triggering an alert through AWS SNS. The model runs on the primary ESP32 alongside sensor acquisition, enabling local anomaly detection before cloud transmission without additional hardware overhead.

VII. RESULTS AND PERFORMANCE ANALYSIS

A. Energy Management and System Performance

All five FSM states were tested throughout the entire operating cycle of the prototype. The transitions from one mode to the next fired at their specified threshold: Mode 3 was activated when the SOC floor reached 20%, Mode 4 was activated when the SOC ceiling reached 80%, and the transition from Mode 1 to Mode 2 was caused by decreased availability of solar energy. There was no manual intervention required for any of the transitions that occurred. During the transition from one relay to another (in the BSS and DCFC), there was no change in the load voltage. There were no interruptions in supply when measuring continuity across all transition points during these transitions.

The measured renewable energy contribution for the prototype was 86.2% compared to the simulated 88.6%, resulting in a 2.4 percentage point difference that can be attributed to converter losses and sensor tolerances; a simulation assumes ideal switching elements with no losses due to conduction so some deviation is expected. The nominated threshold levels and size

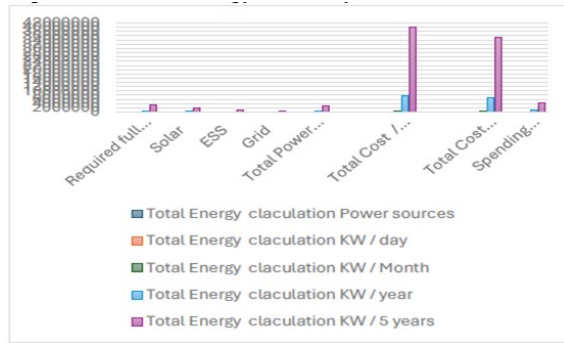


Fig. 16. Five-year energy contribution projections by source showing cumulative Solar, ESS, and Grid contributions against total demand at daily, monthly, yearly, and five-year timescales.

of energy settings transferred directly from simulation to actual hardware with no adjustments necessary. Grid dependency was measured at 11.5% of total energy demand, with the ESS providing the intermediate balance. ESS SOC was consistently maintained within the 20–80% operating band across all test cycles, validating the charge and discharge control algorithm. Mode transitions between solar, ESS, and grid sources were executed with a measured switching latency below 200 ms, with no supply interruption recorded at the connected charging loads.

B. Long-Term Energy and Cost Projections

Fig. 16 provides an estimated energy projection for a full-scale system for the entire five years of operation. It is estimated that the renewable sources will provide about 2.93 million kWh and will represent approximately 88.5% of the total energy needs during the five-year period. Based on the same proportional dispatch ratios that were recorded for hardware validation, the contribution from the grid also shows limited contributions during this entire estimating period. The five-year projection corroborates that the renewable utilization performance from the prototype scales proportionally when under full deployment conditions.

The economic analysis shown in Fig. 17 indicates strong financial viability. The scale of the full-system deployment financial model confirms that the proposed architecture is economically feasible. Daily renewable generation meets most of the energy requirements for the station, with grid usage only taking place for the backup supply during the hardware testing. Revenue from DC fast charging and battery swapping services alone exceeds the daily expenditure on energy supplied by the grid; therefore, a positive net operating position for the station will occur from day one.

The fiscal projection shows that the system is expected to generate approximately Rs. 49.25 Million of cumulative profit over five years, reflecting a cost savings efficiency of 88.5% through improved use of renewable energy sources. The projected annual net profit per station configuration is approximately Rs. 9.85 Million. The cumulative profit from the station over the next five years supports the business case for large scale deployment of the architecture and the financial success of the project will mainly be based upon the

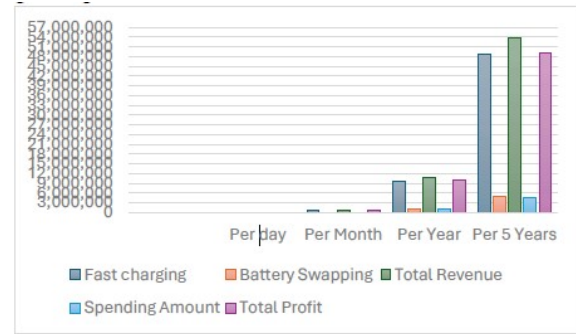


Fig. 17. Five-year cost and revenue projections showing stable grid operational costs against growing fast-charging and battery-swapping revenues.

TABLE I
PERFORMANCE METRICS COMPARISON (SIMULATION VS. HARDWARE)

Metric	Simulation	Hardware
Renewable Energy Utilization	88.6%	86.2%
ESS Round-Trip Efficiency	92.0%	89.5%
DC Bus Voltage Regulation	$\pm 2.1\%$	$\pm 3.5\%$
PCC Transition Time	<150 ms	<200 ms

operational efficiency of using renewables instead of based upon tariff conditions—therefore, this will be a solid business case for large scale deployment of the architecture regardless of the prices charged for energy supplied by the grids.

C. Comparative Performance Metrics

Table I demonstrates how well all four metrics compare between simulation predictions and actual hardware measurements. The close correlations of the four metrics support the contention that the simulated model is a valid representation of the actual physical system behavior and that all four sets of control parameters can be transferred from simulation to the hardware without requiring substantial re-tuning.

Real-world converter losses and tolerances for the sensor readings have resulted in a 2.4% difference between simulation and hardware renewable utilization, which cannot be represented in the ideal switching model. The wider voltage regulation band of the hardware is due to the presence of parasitic impedances of the physical DC bus wiring. All of these discrepancies remain within acceptable limits for a laboratory-scale prototype and do not detract from the validity of the control architecture.

D. Economic Viability Analysis

Scaled as per full-system deployment parameters, the financial model confirms that the proposed EV charging infrastructure is economically viable. Operational analysis has provided evidence to support detailed daily performance indicators. The system generates approximately 1608 kWh per day from renewable sources and utilizes only an average of 209 kW from the electrical grid for the same time period. Therefore, the daily cost of using the grid will be about Rs. 2508 and

there will be an avoidance of costs associated with the grid from using renewable energy of Rs. 19,296 per day.

Collectively, daily revenues for DC fast charging and battery-swap services yield an average of Rs. 29,494 in total daily revenue per station or, on average, a net daily profit of approximately Rs. 26,986. These findings strongly support the claim that the proposed hybrid energy management system provides an economically feasible solution to promoting sustainability through smart charging while utilizing a hybrid method of energy management through both battery-swap stations and DC fast charging stations.

VIII. CONCLUSION

An IoT-based HEMS prototype has been created, tested, and verified under real-world circumstances. The new system allows for coordination of PV energy, ESS battery, and grid power, with 86.2% of the renewable sources seen in operation, closely matching the original simulation data of 88.6%. The 2.4 percentage point difference comes from converter conduction losses and sensor tolerances that the ideal switching model does not capture. That gap was anticipated; it does not change the conclusion that the control logic, threshold settings, and energy sizing transfer from simulation to hardware without re-tuning.

The system operated consistently: the battery's SOC (state of charge) was controlled within the 20–80% band across all test cycles—no deep discharge, no overcharge. DC bus voltage stayed within $\pm 3.5\%$ of 430 V nominal through load steps and source switching events including the 50 kW DCFC initiation surge. Mode transitions completed in under 200 ms at every threshold crossing, with no supply gap recorded at the BSS or DCFC ports. The prototype handled both load types concurrently under the same dispatch framework, demonstrating that architecture is flexible enough to simultaneously serve the load of BSS charging as well as DC fast charging.

The ACFCN anomaly detection layer, operating without labeled training data, demonstrated practical edge-deployability for real-time sensor stream classification directly on the ESP32 hardware. The day-ahead solar forecasting module further showed that predictive alignment of ESS pre-charging with solar availability peaks can reduce grid dependence during demand windows, reinforcing the operational strategy validated in hardware. Net profit projections based on a full-scale implementation amount to a net daily profit of Rs. 26,986.

The system requires no utility-side infrastructure changes, which was a binding practical constraint in the design. What comes next is harder: V2G integration requires bidirectional grid certification that laboratory prototypes do not need, tariff-responsive scheduling requires real utility pricing data, and multi-station coordination introduces communication dependencies that single-station validation cannot expose. The results have confirmed that controls found in simulations can be easily implemented in the physical world, and additional work could provide improvements in terms of V2G capabilities, tariff optimization, and multi-station coordination, among others. The transition from simulation to a functional hardware

prototype provides a validated, scalable resource to develop a sustainable and viable EV charging network.

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